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TITLE PAGE

- **Problem Statement ID –** 1649
- **Problem Statement Title-** DDoS Protection System for Cloud: Architecture and Tool
- **Theme-** Blockchain & Cybersecurity
- **PS Category-** Software
- **Team ID-**
- **Team Name-** HackWeisers

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IDEA TITLE

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IDEA/SOLUTION:-

Implementation of Automatic Protection from DDoS Attack With High Availability Server and Geo-Location

- **Automated real-time testing** for detecting and protection against DDoS attack, Phishing attacks.
- **Alert system** for DDoS attack .
- Secure users using **HTTPS** and secure cookies
- **Redis** for tracking requests in real-time
- **Seamless** integration with already deployed site.
- Using **High Availability** server for data redundancy.

PROBLEM RESOLUTION:-

- Automated testing for abnormal traffic detection and geo-tracking and blocking
- Our solution prioritizes security, high availability and automated recovery.

UNIQUE VALUE PROPOSITIONS (UVP):-

- **DNS Security(DNSSEC):-**Protect against DNS poisoning attacks.
- **Session Hijacking Protection:-** Secure users sessions using HTTPS and secure cookies

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TECHNICAL APPROACH

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TECH STACK:-

Tools used:-

GeoLite2- MaxMind database to access IP information

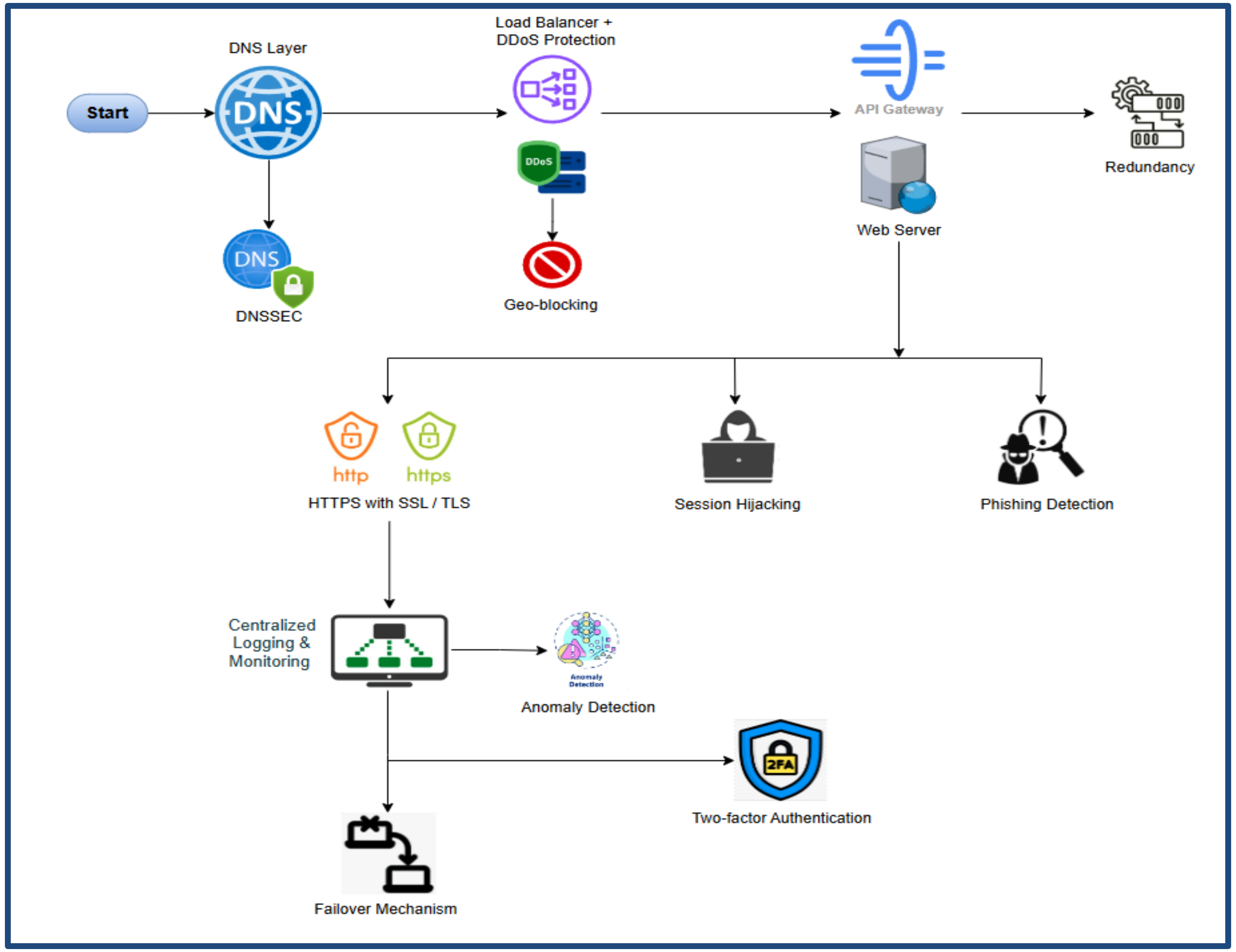
Flask – For blocking blocking specific region generating excessive requests

Redis – For tracking requests in real – time

Bleach – For validating all user inputs before they are used or rendered

Using **Python** to verify the SSL certificate of a website and check for HTTPS

PROCESS FLOW ARCHITECTURE:-



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FEASIBILITY AND VIABILITY

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FEASIBILITY ANALYSIS:-

Technical and Economical Feasibility:-

- Supported by mature technologies, especially in cloud environments. Key challenges include ensuring proper configurations for high availability and security.
- Initial investment is required but justified for organizations needing high uptime and strong security. Cloud-based pay-as-you-go models help reduce costs over time.

Legal and Schedule Feasibility:-

- Compliant with global regulations like GDPR and HIPAA with the correct security measures in place.
- Achievable within a reasonable timeframe, particularly with experienced teams.

CHALLENGES:-

- **Security and Availability**: Challenges include configuring DNSSEC, DDoS protection, and ensuring high availability across load-balanced, geographically distributed servers
- **Authentication and User Experience**: Implementing MFA and OAuth securely while maintaining a smooth user experience is difficult.
- **Scalability and Cost Management**: Scaling centralized logging, ensuring redundancy, and controlling costs are key challenges in maintaining a robust system.

STRATEGIES:-

- **Automate Security and Optimize Load Balancing**: Use automated tools for DNSSEC, SSL/TLS, and DDoS protection, while ensuring session persistence and geographic failover for seamless performance.
- **Scale Authentication and Logging**: Implement cloud-based MFA for secure authentication and scalable logging solutions to handle growing data efficiently and cost-effectively.

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IMPACT AND BENEFITS

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IMPACTS:-

- **<u>Positive Impacts</u>:-**Enhanced security and high availability through advanced measures and cloud scalability boost data safety, system reliability, and flexibility.

- **<u>Negative Impacts</u>:-**Complexity, upfront costs, and ongoing operational efforts can be high, with potential user experience friction from security measures like 2FA.

BENEFITS:-

- **<u>Social</u>**:- Social Benefit: Enhanced trust and safety for users through advanced security measures like encryption, authentication, and phishing detection..

- **<u>Economic</u>**:-Significant cost savings by preventing cyber-attacks, minimizing downtime, and ensuring regulatory compliance.

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RESEARCH AND REFERENCES

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RESEARCH AND REFERENCES:-

❖ **IEEE Xplore Digital Library :-**

- *Journal:* IEEE Transactions on Cloud Computing
- Example article: *"Design and Implementation of High-Availability Architecture for Cloud Services"*..

❖ **ACM Digital Library:-**

- *Journal:* ACM Computing Surveys
- Example article: *"Load Balancing in Distributed Systems: A Comprehensive Survey“*

❖ **Journal of Cloud Computing:-** Advances, Systems and Applications**:-**

- *Journal:* Springer
- Example article: *"Cloud Security and Multi-Factor Authentication in Modern Cloud Infrastructures"*.

❖ **Elsevier - Future Generation Computer Systems:-**

- Example article: *"High Availability and Fault-Tolerance Strategies in Cloud Systems: A Review and Future Directions"*.